```
#Prathamesh Lavange B28
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files iris.csv

iris.csv(text/csv) - 3865 bytes, last modified: 2/18/2026 - 100% done

```
Saving iris.csv to iris.csv
```

```
df = pd.read_csv("iris.csv")

print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

```
print("Dataset Shape:")
print(df.shape)
```

```
Dataset Shape:
(150, 5)
```

```
print("Statistical Summary:")
print(df.describe())
```

```
Statistical Summary:
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198667
std        0.828066     0.433594      1.764420     0.763161
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

```
print("Dataset Information:")
df.info()
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
print("Data Types:")
print(df.dtypes)
```

```
Data Types:
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

```
# Convert species column into category
df["species"] = df["species"].astype("category")

print("After converting species column:")
print(df.head())
```

```
After converting species column:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["sepal_length", "sepal_width", "petal_length", "petal_width"]] = scaler.fit_transform(
    df[["sepal_length", "sepal_width", "petal_length", "petal_width"]]
)

df = df.round(2)

print("After Min-Max Scaling:")
print(df.head())
```

```
After Min-Max Scaling:
   sepal_length  sepal_width  petal_length  petal_width species
0          0.22         0.62          0.07         0.04  setosa
1          0.17         0.42          0.07         0.04  setosa
2          0.11         0.50          0.05         0.04  setosa
3          0.08         0.46          0.08         0.04  setosa
4          0.19         0.67          0.07         0.04  setosa
```

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df["species_encoded"] = le.fit_transform(df["species"])

print("After Label Encoding:")
print(df.head())
```

```
After Label Encoding:
   sepal_length  sepal_width  petal_length  petal_width species  \
0          0.22         0.62          0.07         0.04  setosa
1          0.17         0.42          0.07         0.04  setosa
2          0.11         0.50          0.05         0.04  setosa
3          0.08         0.46          0.08         0.04  setosa
4          0.19         0.67          0.07         0.04  setosa

   species_encoded
0                0
1                0
2                0
3                0
4                0
```